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Analysis and the rationalisation of production

THINK OF AN EARLY TWENTIETH-CENTURY chemistry laboratory – a big, airy room with rows and rows of bottles full of chemicals. Is it a museum of minerals, of plant and animal products and of the creations of industry? In a way, perhaps, but for the most part, students of chemistry do not enumerate the specimens, worry about the arrangement, look for gaps in the collections or prize new acquisitions, as they might indeed have done in the eighteenth century. Why not? Partly because, since the early nineteenth century, they could buy specimens from companies set up for that purpose, but more fundamentally perhaps, because the chemicals seem to us somewhat ‘arbitrary’ and lacking in individuality. They are almost all ‘compounds’, which we can manipulate into other compounds, provided they contain the correct elements. For most purposes each chemical is adequately specified by naming the elements contained and their ratios. We do not want to know how the specimen was obtained – whether from nature or industry. That would only matter if we were worried about the traces of ‘impurities’ that ‘distanced’ our sample from being simply what it is supposed to be – a single compound of a few chemical elements. To put it briefly – these chemicals are no longer known through natural history; in the laboratory they are the creatures of chemical analysis.

This chapter and the next trace the formation of analytical sciences, especially around 1800; I stress their relations with professional education and with ‘consultancy’. With ‘analysis’, we approach the knowledges which are esoteric and therefore seen as characteristic of Western science, technology and medicine.

We leave the realms of meaning and those of natural history, and move into expert knowledges that are less continuous with everyday knowledges. In Britain, this kind of science is said to be *not* 'just common sense'. And in the French tradition of history of science associated with Gaston Bachelard, the analytical sciences stand on the far side of the 'epistemological obstacles' which are said to have barred the way to true science, pushing us back into the realms of the everyday with which we are familiar and comfortable. To use one of Bachelard's key examples: the experience of fire was (once) a key part of our phenomenological world; it was freighted with symbolic meaning and literary associations; it was an object of hermeneutics as well as natural history. To talk about fire as the combination of oxygen with other elements, or about the release of specific quantities of energy, was to break from the mundane and enter a world beyond common experience and associations (Bachelard, 1938; Gutting, 1990).

That we can see chemicals in this way is the achievement of the late eighteenth century, though it was not without precedent. This chapter and the next argue that the same historical period created many other analytical sciences, each with their own elements. The elements were not obvious – they needed to be discovered or perhaps invented. I shall argue that Newtonian mechanics, the creation of the seventeenth century, served as a key model; that some new analytical disciplines, notably chemistry, then served as models for others; and that, especially in France, the pioneers of the new developments were self-conscious about method. The key period, c.1780–1850, is sometimes described by political historians as the Age of Revolutions – for the French Revolution which began in 1789 and echoed for half a century, and for the Industrial Revolution that spread from Britain. By extending some suggestions in the masterly works of Charles Gillispie, we might also call it the 'age of analysis' (Gillispie, esp. 1965; 1980), and that can be our clue in trying to *explain* the developments.

We shall see that engineers systematically decomposed machines into elementary machines, that other engineers and natural philosophers found how to trace 'mechanical action' through machines, and that engines came to be seen as systems

through which the element heat flowed. I shall argue in this chapter that much of what later became physics appeared in the early 1800s at the edges of chemistry and of engineering, as studies of 'elements' such as light, heat, magnetism and voltaic electricity. I include as analysis both the decomposition of 'compounds' into their various elements, and the reduction of systems to the 'flow' of single elements. Analytical chemistry was the model of the first type, thermodynamics could serve as a model of the second. At the end of this chapter, I explore the links between analysis and industrialisation, by considering the 'rationalisation' of production and by musing on the relationships between analytical elements of the new sciences and the 'elements' then apparent in technologies.

In the next chapter I show how animal and human bodies were deconstructed into elements – tissues and then cells – how plants were similarly decomposed, and how these developments were related to changes in medical education and practice. I consider how *form* was analysed in the new 'morphological' sciences, especially in German universities, and how geology became the study of strata (stratigraphy). I also explore the roles of analysis in the new social sciences. But we must not forget the continuing importance of the non-analytical forms, or the *interplays* between analytical knowledges and the 'natural-historical' and/or 'hermeneutic'. Those interplays were central to the politics of science in the age of analysis and they remained so, as I shall try to show.

This chapter and the next focus on the period 1780–1850, when this raft of new analytical sciences and technologies were discovered/constructed. I present them, in large measure, as products of major *institutional* novelties: French professional schools, hospitals and museums, German reformed universities, and the British Industrial Revolution. At the end of Chapter 5, I explore the extent to which institutional and industrial change (and hence wider political and economic factors) may be said to *explain* conceptual change (and vice versa). But I also want to emphasise the importance of later and earlier examples; we continue to create new analytical disciplines – e.g. geonomics. By contrast, some analytical disciplines were constructed in the

ancient world and revised in the period we call the scientific revolution. These classical disciplines were about motion – the movements of the planets and, later, the motions of matter on earth. With them I begin.

Analysis from the ancient world

In the chapters on readings and on natural history we mentioned the *meanings* of Greek cosmologies and the Aristotelian tradition of natural history but, in at least one respect, Greek cosmologies drew on other knowledge traditions with very different aims and accomplishments. The Babylonians had recorded the movements of planets and stars in numerical terms – for predictions, not least for astrology. It was Eudoxus, a pupil of Plato, who attempted to draw Babylonian calculations into Greek cosmologies by interpreting numerical ratios in geometrical terms – expressing intervals as circular movements. This mixed tradition was developed by Greek colonists in Egypt. There, in Alexandria, Ptolemy and his school elaborated this line of investigation in the form later transmitted to the Christian West. The movements of the stars and planets could be modelled by a system of about ninety spheres circling round the earth and round each other (Kuhn, 1957).

This way of doing science might usefully be called analytical: skilled, esoteric practitioners (here mathematicians) successfully ‘reduced’ the baffling motions of the planets to a system of circular movements. The system was not merely descriptive, it predicted motions for calendar-making and astrology. Whether this system was ‘real’ or whether it was to be regarded as a calculating device to ‘save the appearances’ became a matter of disputation among medieval and Renaissance scholars; that was part of the discussion around the radical revision of the Ptolemaic scheme suggested by Nicholas Copernicus in 1543, when he showed that a more elegant model could be obtained by putting the sun at the centre of the universe and making the earth into one of its planets.

Between 1543 and about 1700, as noted in Chapter 2, the common Western model of the cosmos was transformed. An Aristotelian model of a closed world gave place to an infinite universe; a complex apparatus of cycles and epicycles was replaced with elliptical orbits around the sun. There was no longer a deep distinction between the celestial and the mundane, and in Newton's great synthesis, celestial motion obeyed the same laws as motion of bodies upon earth. By 1700, in principle, matter in motion obeyed simple laws, wherever and whenever, and this heroic model of analysis set the aspirations for many other realms of knowledge, e.g. chemistry, which were then compounded of natural history and craft, underlain with hypothetical universal explanations in terms of mechanisms or of the Greek elements – earth, air, fire and water.

So early modern 'analytical instruments' might perhaps be seen as models of the world which were also tools for living. Clocks we have met already, as models of the heavens that became regulators of men; perhaps we can add the instruments of the surveyor and the navigator, plus aids to calculation. Perhaps we should see them as the tools of the trades which analysed the earth and the heavens (Bennett, 1986), marking out intervals of time and space. Displayed in an eighteenth-century 'cabinet', they could be said to represent mathematical analysis just as the rows of specimens represented the 'history' of nature and of artefacts, and as the chemists' 'instruments of torture' represented the 'experimental histories' of the Baconian tradition. The practical uses (and prestige value) of the navigational and mathematical instruments ensured a supply and a skill base which could be drawn on by teachers or travelling lecturers who wanted variant instruments or more sophisticated versions (Stewart, 1992).

Analytical aspirations

Those who 'mechanised the world picture' in the seventeenth century separated the primary qualities of matter and motion

from secondary qualities such as colour. In our terms, they separated the 'elements' of a successful analytical programme from those qualities which seemed less tractable, not least because they were hard to quantify. They argued that only matter and motion were sufficiently clear and distinct to be fundamental to understanding, that here was the deep accord between God and man; God made clocks of infinite quality.¹ Mechanical models were speculatively extended to chemistry and to anatomy and medicine, and they served to stimulate observational studies; several key microscopists searched the inner structures of plants and animals hoping to find the 'looms' on which the tissues were woven – but the results were meagre compared to celestial mechanics and the programme of 'mechanisation' was failing by c.1750 (Schofield, 1969).

But as that programme was fading, another was strengthening, which also drew on the successes of celestial mechanics and especially on the fame of Newton. Instead of emphasising the mechanical constitution of the world, it emphasised the simplicity of the laws to which apparently complex phenomena could be reduced. This approach, which developed into the philosophies of science known as positivism, was characteristic of the late 'Enlightenment'. Ernst Cassirer, in his account of the period, stressed how the laws of motion served as paradigms of explanation, and how procedures such as the 'resolution of forces' served as models of *analysis*. The complex motion of a cannon ball could be resolved into horizontal and vertical components, and thus expressed as a compound of simple motions (Cassirer, 1955). That model may serve as a simple example of the ambition to analysis which by 1850 had been successfully generalised across many new disciplines – not by reducing biology or medicine to matter in motion, but by finding/creating 'elements' by which new disciplines could be constituted.²

Michel Foucault, in his early works, drew attention to a major fault line in the 'archaeology of knowledge' c.1800; he explored it for medicine, life sciences, economics and language studies, but he excluded physical sciences (Foucault, 1970; 1972; Gutting, 1989). Thomas Kuhn, in his essay on mathematical and experi-

mental traditions (Kuhn, 1977b), argued that many new physical sciences took their modern forms at this time; studies of heat, light, electricity and force emerged in mathematical form, alongside the 'reformed classical sciences' such as mechanics. By combining and regrounding the accounts of Foucault and Kuhn, I want to trace the 'displacement' across *most* of STM, including chemistry, engineering and the new physical sciences, as well as medicine, biology, earth sciences and some new social sciences. For many of these new sciences, the new chemistry was exemplary.

The elements of chemistry

We have suggested that eighteenth-century chemistry, like most other Baconian sciences, was a mix of natural history and craft, together with such natural philosophy as might provide speculative explanations of the phenomena. Practical men had continued to advance the chemical arts of medicine, agriculture and manufacturing, as they had done since the Renaissance; savants had tried to effect classifications, modelled on those in botany, or to give explanations of chemical properties based on the Greek elements (earth, air, fire and water) and/or the alchemical principles. But by 1789 the French academician Antoine Lavoisier was offering his *Elements of Chemistry* – a textbook of a new system of chemistry, written in a 'new language'.³ He drew on new discoveries and new perspectives concerning 'airs'. Common air was no longer a certain quality of air or a blend of qualities, such as the ability to support fire; it was understood as a mixture of various different *kinds* of air, notably those that came to be called oxygen and nitrogen. These could be separately prepared, and had constant qualities, as did other kinds of gas such as hydrogen. Lavoisier allowed these new gases to be elements if they resisted attempts at decomposition. They could be weighed, as could other substances such as metals, which also resisted decomposition. Decomposable substances could then be understood as compounds, and they could be named accordingly.

That this new scheme was worked out in France may be due to the confluence there of the *various* traditions which proved crucial – an interest in the taxonomy of chemical stuffs, information about new gases and reactions, the attraction to quantification and weighing, the widespread concern with ‘analytical’ understandings and a ‘pre-positivist’ readiness to define ‘elements’ in a way which was operational and specific to each new discipline. One can see this methodological and pedagogical programme in the presentation of Lavoisier’s work – in a textbook meant to create disciples who would ransack the old chemistry for all they could translate into their new language (Donovan, 1996; Holmes, 1985).

Though Lavoisier and other (salaried) members of the Royal Academy were often called upon to advise the government on technical questions – sometimes at the royal manufactories for munitions, tapestries and porcelain – it seems unlikely that his ‘new chemistry’ can be seen as deriving substantially from these roles. But after Lavoisier was condemned as a tax farmer and executed on the (new, rational) guillotine, his chemistry was substantially developed by the state’s mobilisation of chemists during the wars with the rest of Europe that extended almost without break to 1815. For two decades, France had to be self-sufficient economically and technically while supporting a huge conscript army.⁴ We shall see the importance of this conjuncture for engineering and for medicine; for chemistry the effects were perhaps less clear because there were no big teaching institutions dedicated to the subject. For the most part, a group of leading chemists and natural philosophers, some of whom had private laboratories, served as consultants and directors in government workshops and factories. By 1815, they at least were sure that savants in general, and especially the new chemists, had been of huge benefit to the state (Alder, 1997; Crosland, 1967; Fox and Guagnini, 1998: 71–4; Smith, 1979).

The new chemistry led quite quickly to new models of the physical world. The Manchester teacher, John Dalton, soon saw that different elements might have atoms of different weights, and that in compounds (as opposed to mixtures) atoms of dif-

ferent elements might be combined (Cardwell, 1968; Patterson, 1970; Thackray, 1970). The Swedish chemist, J. J. Berzelius, teaching medical chemistry and mineralogy in Stockholm, regularised the new system and introduced the formulae we still know (Melhado, 1981; 1992). The 'operational' potential of this new paradigm was enormous: all the substances in the world waited to be analysed into the new elements, and all the chemical changes in the world awaited description as reactions between elements.

In France and in Sweden, the new chemists were often employed in state institutions for medicine, pharmacy and mineralogy. In Britain, as we shall see, much of the demand came from industry, and chemists acted as 'consultants'. In all these countries, perhaps especially in the USA, agricultural analysis – especially of soils and fertilisers – became a major preoccupation by mid-century. But the key country for the production and reproduction of the new chemistry was Germany, and especially the institute which Justus Liebig developed at the University of Giessen in the 1820s, and which educated so many of the academic and professional analysts of the nineteenth century (Brock, 1997). In that small university town, Liebig successfully institutionalised chemical analysis by drawing on three key factors. He learned the methods in Paris, though the opportunities there for serious students were limited; he applied them in the education of pharmacists – a very important financial support for several key chemists *c.*1800; and he adapted to laboratory sciences the ideology of collective research cum advanced training which was the pride of the newly founded (or reformed) German universities.

'Research' as the chief aim of universities was then a new idea. As we shall see later, its early practitioners were not physical scientists but philologists who analysed ancient texts, or mathematicians (Olesko, 1988). The German universities reformed or founded after the French occupation, stressed language and *culture* as common features of the many 'German' states, sometimes contrasting them with French analysis. While the French stressed professional education, the Germans stressed broad

'learning' in a community of teachers and scholars – designed to cultivate both knowledge and men (Shaffer, 1990; Turner, 1971). Liebig managed to *combine* the creation of new knowledge with the training of professionals, and by the 1840s he was the acknowledged master of European chemistry. As younger peers and protégés pushed on to new theoretical problems – e.g. how to effect new classifications of compounds, Liebig built up his long-standing links with the world of practice. He published books on the uses of chemistry in agriculture, medicine and industry (Turner, 1982), and his students became consultants in these fields. For the most part their work was analytical – this chemical programme and its progeny probably comprised the nineteenth century's largest investment in laboratory science.

It would not be difficult to show that the age-old interplay between 'chemistry', medicine and the industrial arts had been transformed by the new understandings and practices of analysis. Or that state support for elite chemists in France, for universities in Germany, and for such common national enterprises as mining and the manufacture of gunpowder, all provided resources and incentives for the new chemists. In Britain they depended chiefly on the Scottish universities and the private markets for education and especially for technical advice. By the mid-nineteenth century, British analytical chemists could make a living doing consultancy work in their private laboratories, and maybe teaching part time in the local medical school or 'mechanics' institute' (Russell, *et al.*, 1977; and see Russell, 1996). That the reform of *pharmaceutical* education and practice seems to have been particularly important in several countries (Hufbauer, 1982) parallels the emergence of other forms of analysis in the context of higher professional education. This is clear for engineering, as for medicine, especially in France.

Analysis for engineers

Engineering and mathematics have been related since ancient times. Geometry was a tool for surveyors as well as a clue to the

ideal form of the cosmos. If terrestrial mechanics was created in the Renaissance to join celestial mechanics as an analytical science, it was in part because of the interaction between scholars and practical men responsible for fortifications, ballistics, etc. But as in other areas of STM, the reach of analytical methods was limited, and much engineering knowledge continued at the level of natural history and craft recipes.

When, in the mid-eighteenth century, the French government established a school for military engineers, there was no agreed curriculum and no set way of teaching. As Charles Gillispie showed so well, the college was a kind of collective apprenticeship, including mathematics and mechanics. But gradually the teachers took charge of the educational programme, establishing a syllabus of courses which built on previous courses, introducing classroom exercises to supplement lectures, together with examinations to test the student's mastery of each course (rather than just serving for the award of prizes). All these pedagogical devices seem so familiar to us that we need reminding of a world where they had not existed, and of the circumstances of their invention (Gillispie, 1980).

These initiatives in engineering education were pushed by young lecturers keen to make careers and reputations – to produce good engineers, but also to produce scientific work for their own prestige. Textbooks would serve their own students, but they would also be used by other engineers, inside and outside training institutions; they could serve as scientific treatises, but also for 'learning at a distance' (Picon, 1992; 1996; Weiss, 1982). It was in this context that Gaspard Monge and others developed a series of methods that would be central to engineering education down to the present. They extended Renaissance insights into analytical disciplines, including the 'projective geometry' which later generations would learn as 'technical drawing'. Such drawings became crucial to engineering practice, but they also had another significance for teaching institutions – they helped bring a wider range of machines and constructions into college. For much the same reasons, such colleges, like medical schools, developed *collections* of machines;

museums became a part of this practical professional education. For big machines and civil constructions such as bridges, scale models were used – to allow vicarious experience without travel, to facilitate comparisons, and to help make clear the key features of the machine in question.

But if machines were collected and compared, what was to be the framework of comparison? Should it be *function*, comparing different kinds of prime mover, say? Such classifications had been used in some contemporary anatomical museums (e.g. that of John Hunter in London) and they were to remain a staple of technological museums. But could you dissect machines (as you could bodies) and find their elements – the ‘simple machines’ from which all others were made? An interest in this question was clear in Renaissance mechanics: Galileo wrote about the wheel, the lever, the screw, etc. as simple constituents of more complex machines; but in engineering, as in medicine, it was in the decades around 1800 that the first systematic texts appeared, breaking down machines into elementary parts and presenting the resultant classifications of machines as an authoritative form of engineering understanding.

The authors of the text regarded as seminal were Jean-Marie Lanz, a young Frenchman, and Agustin de Betancourt, a Spaniard who had supervised the construction of hundreds of models intended for a new technical college in Madrid. I do not know the extent to which Betancourt drew on other contemporary STM, or on the contemporary vogue for analysis as a general method, but such comparisons were explicit for one of the founders of advanced technical education in German states (Perez and Tascon, 1991). Von Reuleaux, in the mid-nineteenth century, drew parallels between his analyses of machines and comparative anatomy; he drew a second parallel with chemistry, and was so led to devise a nomenclature for ‘elements of machines’ by analogy with the agreed formulas for chemical elements. As in chemistry, one function of this nomenclature was to devise *possible* compounds that might be *synthesised* (Reuleaux, 1876).

By then, analysis of machines (and projective geometry) had

become mainstays of German technical education. They were particularly useful for pedagogues who wished to assert that higher technical education was practical but also had the rigour expected of traditional *university* disciplines. In some such colleges, machine elements were available in standardised form, to be assembled by students who thereby developed the 'analytical eye'.

Analysis and 'physics'

Historians of physics have tended to see nineteenth-century physics as an extension of natural philosophy and mathematics rather than of the new chemistry or engineering, but perspectives are changing. It would be worth checking how many of the nineteenth-century experimental physicists had trained as chemists.⁵ Donald Cardwell and his Manchester associates showed that 'energy' studies in Britain owed much to engineers (Cardwell, 1971; 1989; Pacey, 1974), and Crosbie Smith and Norton Wise have continued the lesson more recently by investigating the career of William Thomson, Lord Kelvin, both in Glasgow University and among the industrialists and marine engineers of that Second City of Empire (Smith and Wise, 1989). And it was graduates of the Ecole Polytechnique, the school for military engineers established in 1793 after the Revolution, who *made* French physics c.1830 (Fox, 1992; Geison, 1984; Gillispie, 1980; Shinn, 1980). Perhaps we should take the engineering tradition more seriously – not just for the analysis of machines into components, but for the analysis of 'flows' of 'elements' through machines and through 'nature'. The elements here were 'mechanical action', 'force', 'heat' or, later, 'energy'; the central machine was the steam engine.⁶

We know little about the invention of the first practical steam engine, devised by Thomas Newcomen to pump water from Cornish tin mines. Later eighteenth-century improvements, especially the separate condenser, were made by James Watt, an instrument-maker in Glasgow University who worked for James

Black and knew his studies on heat. Black, by elucidating 'latent heat' and in separating 'quantity of heat' from temperature, had greatly advanced the analysis of heat and this was of advantage to Watt (Basalla, 1988; Cardwell, 1971; Donovan, 1975; Hills, 1989; Jacob, 1997).

It was Lazare Carnot, one of the founders of the Ecole Polytechnique, who provided the foundational analysis of 'mechanical action' in machines. The question at issue was how 'force' could be transmitted most efficiently through a set of cogs and levers (say). Carnot showed that the transmission was most effective when it was smooth, without jarring of the parts; in his *Essai sur les machines en générale*, he focused on the 'work' done by machines (Gillispie, 1971). At about the same time, the British engineers who were emerging as key players in the Industrial Revolution stressed 'mechanical power' or 'duty' as a measure of what the new steam engines could do. Thereafter we can trace the two 'national traditions' to the energy doctrines of the mid-nineteenth century. In Britain, practical engineers and their philosophical associates reflected on the potential of engines of various kinds; formal teaching of engineers was scarcely an issue until after mid-century (Buchanan, 1989; Cardwell, 1957). In France a series of investigators, schooled in mathematics and the analytical techniques of the Ecole Polytechnique, created new disciplines expressed in textbooks and in papers to the Academy of Sciences in Paris (which was *not* a major centre of industry).

In France, the generation after Lazare Carnot, including his nephew Sadi Carnot, created mathematical analyses of heat flow in machines. They used the mathematics developed by Joseph Fourier, a graduate of the Ecole Polytechnique who had rejected the 'microphysical' models by which Laplace and his colleagues in the Academy of Sciences tried to extend Newtonian analysis into a general theory of attraction between all particles of matter. By contrast, Fourier's 'theory of heat' was essentially macroscopic, geometrical and practical – the engineer's view from *outside* the machine, an assessment of what a machine could *do* (Grattan-Guinness, 1981; Grattan-Guinness and Ravetz, 1972; Smith, 1990: 328).

By contrast, in early nineteenth-century England, there was little professional education in engineering (or in chemistry). The technical and philosophical analyses took place chiefly in the 'societies' of Scotland, London or Manchester, where practitioners with a taste for analysis, and teachers oriented to practice, interacted informally and through publications. Engineers who made and operated steam engines concerned themselves with their 'duty', or horse power – their capacity to drive pumps or machinery and therefore their economic efficiency (Morrell and Thackray, 1981; Smith, 1998). In industrial Manchester, where 'duty' was well known, we can trace these concerns through to the 1840s and the work of James Prescott Joule who extended considerations of efficiency to the new electric motors, and to whom we owe the principle of the mechanical equivalent of heat.

By trade, Joule was a brewer, by inclination a cautious investigator; politically, he was a Conservative in a Liberal-radical town. He had been raised to science as a pupil of John Dalton, who had achieved world fame for interpreting French chemical elements in terms of atoms. By contact with Dalton's friends, and at meetings of the Manchester Literary and Philosophical Society, Joule became familiar with problems of engine efficiency and with the tradition of 'modelling' machines to investigate such issues. His experiments which became classic were provoked by popular enthusiasm for 'electric motors', touted in the 1830s as capable of indefinite speed and power output. Joule was sceptical. He set up model electric motors to measure the 'work' produced by a given amount of zinc in the battery. (All this technology was *very* new.) He concluded that electric power was, in fact, rather expensive compared to the benefits of burning coal in a steam engine. He went on to measure the relationship of mechanical action to heat production, using a 'paddle wheel' immersed in water heated by its rotation. In all these experiments Joule was measuring relationships, assuming that 'force' or 'action' could not be destroyed, but only converted into some other form (Cardwell, 1989).

The two traditions came together in Britain when reformers of 'natural philosophy' imported French methods. At Cambridge,

the capital of British mathematics, French methods were central to a revival movement *c.* 1830, led by young graduates interested in engineering and industry. Charles Babbage, later a 'philosopher of industry' and the inventor of a proto-computer, was prominent, but it was in Glasgow that the new mathematics chiefly met the new engineering. The only British industrial city which then had a university, created a professorship in engineering in 1840, and six years later they appointed to the chair of natural philosophy, the young mathematician and experimenter, William Thomson, later Lord Kelvin. William's father was a teacher of mathematics, his brother James was a pupil of the Glasgow engineering professor and then an engineer in a London shipyard and marine engineering works. William had studied mathematics at Cambridge and spent some time working with Victor Regnault in Paris, a chemist turned physical experimenter who taught advanced measurement techniques and conducted extensive researches on the relation between pressure, temperature and specific heats; his data was simplified into 'tables' for engineers – a characteristic (analytical) guide to practice. Appointed to Glasgow at the age of twenty-two, Thomson also focused on the analysis of steam engines, and on analogous analyses of electric motors. The work expended or absorbed in an electrical engine depended on the potential difference, just as it depended on the temperature difference between the boiler and condenser of a steam engine, or the height of fall for a water-wheel (Smith, 1990; Smith and Wise, 1989).

This theory about the flow of heat was based on French analyses to which Joule had objected on metaphysical grounds: they seemed to allow the 'annihilation of force' which for Joule was a power belonging to God alone. Joule had proposed that the 'force' was converted into heat. Thomson, by 1851 was able to reconcile those views by the principle of 'conservation of energy'. Though it is well known that other scientists arrived at this principle by other means (Kuhn, 1977a), I emphasise here that the best recent accounts of the key French and British traditions demonstrate the plausibility of seeing 'energy physics' as based in engineering analyses; the principles of nature could be derived

from machines. But how were manufacturing-machines made and understood; what kinds of analysis helped *produce* them? In what ways were the factories of the Industrial Revolution a matter of analysis?

Rationalised production

We return here to the points I made in the introductory chapter to this book, about the relation between ways of knowing and ways of making – especially to the relation between analysis and the rationalisation of manufacturing production. Perhaps we can see the Industrial Revolution as extending the practice of analysis from the divisions of time and space in the early modern period to the divisions of labour in the modern age. Arguably, a mechanical loom was to weaving as a clock had been to the heavenly motions – a materialisation of analysis.

Here we can follow Edmund Cartwright, the clergyman and natural philosopher who invented the mechanical loom in the early years of the Industrial Revolution. For him an automatic loom was not just an enticing economic proposition, it was a philosophical challenge: only if he could mechanise weaving, could he be sure he understood it. By analysing the process into simple human actions, he could substitute mechanical actions that together would reconstitute the process (O'Brien, Griffiths and Hunt, 1996). Thus understood, looms belong to the world not just of clocks but of the automata which had fascinated eighteenth-century high society – those doubly enchanting products of a deconstructive, disenchanting analysis.

But, importantly, 'rationalised production' was not necessarily 'mechanisation' because not all 'machines' were made of metal or wood; they could be made of *people*, treated as mere components. Indeed, it could be argued that the mechanisation of people was historically prior to the construction of inanimate machines. Lewis Mumford, in a seminal article on 'Authoritarian and Democratic Technics' (Mumford, 1964), long ago suggested that the sets of slaves who built the pyramids should be seen as

precursors of mechanised production. Adam Smith's classical analysis of increased productivity in eighteenth-century Britain referred not to new machines but to 'division of labour' in workshops; in pin-making, to quote the famous example in his *The Wealth Of Nations* (1776), the separation of tasks between different workers allowed each task to be performed more efficiently. Here, indeed, was the basis for mechanisation; but in one sense the key step had already been taken – the elements of the productive process had been separated, each had been developed as the sole work of one component, and these components had been assembled to form a productive mechanism which reflected and expanded the work once carried out by a single skilled craftsman (Babbage, 1832; Berg, 1979; 1980; Marx, 1967).

For such reasons, we should use 'rationalised production' (rather than mechanisation), as a basic 'type' of 'making', allowing that the elements were usually then mechanised, more or less. This rationalisation of production, from the industrial revolution factories to the production lines of Henry Ford and the 'ergonomics' and 'scientific management' of Gilbreth and F. W. Taylor, lies at the core of 'industrialisation' over the last two centuries.⁷ (Though here, too, we should remember that old and new crafts were also involved – e.g. crafts of machine-making or of canny maintenance (Braverman, 1974).) Relentless analysis and the standardisation of machines and of men have been central to most dystopic visions of modern manufacture and of modern society more generally.

Rationalisation and identities

But how are we to relate the 'scientific' analysis with which we began this chapter and the 'technological' analysis now under discussion? If Cartwright analysed the action of a weaver, or the English farmer, Robert Bakewell, the desirable properties of a sheep carcass,⁸ they provided frames for systematic discussion and improvement, but they did not thereby establish new 'sci-

ences', as did Lavoisier with his new chemical elements, or Xavier Bichat (as we shall see) with his tissues of the body.

How their contemporaries saw that difference, and how we feel it, deserves more attention than I can devote to the task. Perhaps realist philosophers have already shown how it is that 'elements' became 'real' rather than merely instrumental. The problem is doubled in as much as some elements that seemed philosophically secure, e.g. 'heat' or 'caloric', were later turned into mere 'properties' of more substantial elements – heat was not a 'stuff', but rather the motion of other kinds of molecules. From the end of the nineteenth century, even the familiar chemical elements lost their specificity. In principle, and under peculiar conditions, one could find several kinds of phosphorus, say; each chemical 'element' had become a collection of 'isotopes', a range of possibilities in a matrix of subatomic particles.

This shift from a 'kind' to a 'range' may seem remote from industry but, in fact, it was also a key aspect of rationalisation of production. Here my initial guide was an article which explored understandings of textiles in France from the eighteenth century through the early nineteenth century. In a nice instantiation, corresponding to my general argument about craft-objects and natural-historical species, William Reddy showed how the system of local guild production of textiles required specifications of local cloths. These were printed in books that read like natural histories, but were produced by an inspectorate. Though with the rise of unregulated, extra-guild production these accounts lost relevance, they continued to be produced *faute de mieux* (Reddy, 1986). Only after the onset of industrialisation was a new genre of text printed; and these were about production rather than specific products, about mechanised spinning and weaving, about the capacities and versatility of machines. Of course, information about products continued to be available in France as in Britain, but in the form of catalogues and sample books (the obvious equivalents of floras, or collections of pressed plants). In a sense, the specific products had become points (or groupings) on the grid of possibilities which machines could realise.

We may note the similar contrast between reproducing pedigree animals and the more utilitarian forms of breeding which were designed to secure new sets of characteristics.⁹ In Britain from 1750, cattle and sheep were changed radically as breeders sought marketable characteristics and faster growth. Sheep, Bakewell said, were machines for turning grass into money; they could be redesigned to do so more efficiently. And that new mutability of domestic animals was well known both to Charles Darwin and to his contemporary, Gregor Mendel, the Moravian monk and science teacher who in the 1860s discovered mathematical ratios in the inheritance of plant characteristics. Indeed, both the theory of evolution by natural selection and Mendel's work on inheritance can be shown to have built on this shift in breeding technology (Desmond and Moore, 1992; Orel, 1984).

In all these cases, 'species' – whether manufactured, chemical or biological – lost something of their 'given-ness', and came to appear as points (or clusters) on a matrix. In my terms, they moved from natural history to analysis and rationalised production. But note that pedigree animals, like wines and luxury brands, remained as quasi-species.

Production and analytical sciences

From this discussion of production and products there emerges a picture which complements our account of analytical ways of knowing. We may see 'technological' analyses of productive processes, whether mechanical, biological or human, running alongside 'philosophical' analyses of 'model' situations. 'Technological' here refers to direct analysis of technologies – e.g. by testing waterwheels, measuring by-products of chemical plants, or keeping data on rates of fattening in cattle. In all those cases, practical investigators looked for regularities that could guide understandings and improvements. Statistical trials of drugs in hospitals fall into the same class if conducted on the basis of 'suck it and see' (Matthews, 1995).

More 'philosophical' investigators sought to uncover stable 'elements' behind natural and artificial phenomena – usually looking for them in special circumstances, for example in laboratories where phenomena could be simplified, dissected and measured more easily. The elements included those of the chemists, the tissues of the body (and eventually, Mendel's units of inheritance, which became genes). Such elements could then be used for elucidating productive processes – though often it was difficult to link analysis-in-principle with operations-in-practice, and many industrial processes continued to be operated according to 'technical' rules of thumb – i.e. by natural history and craft.

If, then, this analysis of analysis is persuasive, we may need to rewrite the history of physical sciences so as to give more weight to worldly practices, especially engineering. The historical relation of physics to engineering may indeed come to resemble that between physiology and medicine, or of chemistry and heavy industry. In all cases the 'science' provided a language in which to analyse healthy or unhealthy 'workings'. These languages of analysis were crucial to 'consultants' and to teachers who made a living out of expounding and exemplifying the principles, especially in the institutions of higher professional education in France and Germany. In these 'places of learning', analysis took precedence over craft (Layton, 1971; 1974).

Such analyses were also important to practitioners, especially when there were 'difficulties'. But though consultant analysts provided diagnoses and suggestions, they did not often 'produce new practice'. In the 'places of practice', e.g. factories and English teaching hospitals, practitioners were usually in charge on the basis of 'experience'; analysis, there, was for keen junior staff or consultants called in on special occasions.

In private industry and especially in Britain, graduate chemists worked chiefly as consultants. By analysing the inputs, outputs and by-products of industrial processes they could effect large savings, as Andrew Carnegie was to discover in his steelworks (Mowery and Rosenberg, 1989: 28–34). Many analytical chemists were self-employed, typically they were 'liberal'

scientific professionals (Russell, 1977; 1996). But by the end of the nineteenth century they were also working for government in the enforcement of industrial and public health legislation that they had helped promote. The first inspector of 'pollution' under the British Alkali Act was Robert Angus Smith, a Manchester chemist now best known for discovering 'acid rain'.¹⁰ Chemists also featured as experts in patent disputes. In all these roles – improving production, assessing pollution, devising regulations, etc. – chemists functioned chiefly as *analysts*; and for the most part that is what was taught in institutions of chemical education, including the courses on technological chemistry which began to appear in technical colleges, especially in Germany.

Mechanical and civil engineering also offered opportunities for consultancy – for assessing new bridges or dams, or for standardising boilers (Buchanan, 1989). Some of the work was for the firms concerned, some for government, some for insurance companies. To take one key example, the devastating effects of boiler explosions prompted much expert analysis of the safety requirements, and insurance regulations played a major role in establishing reliable safety standards. Such roles brought mechanical engineers into the business of legislation. Civil engineers had long been involved, advising on schemes for roads, canals, railways and reservoirs, which usually needed parliamentary permission.

We shall deal in later chapters with the growth of 'electrotechnics', but can note here that the massive expansion of electrical industries at the end of the century was based in part on new techniques of electrical analysis and standardisation, on new instruments robust enough to work outside laboratories and on technicians able to diagnose and repair electrical circuits and machinery. Those were the skills taught in the new technical colleges and in the university departments of physics and electrical engineering which mushroomed to meet the demand and/or to benefit from the excitement (Fox and Guagnini, 1999: ch. 3). (Much the same could probably be said of the electronic industry several decades later.) By seeing most chemists and consultant engineers as analysts, we clarify their roles, their social relations

and their common features, and we prepare ourselves to recognise the huge importance of these activities down to our present.

We shall return to analysis and industry in Chapter 7, but we turn now to analysis in medicine and the life sciences, and in the sciences of the earth and of society. Again we backtrack to the seventeenth century, before focusing on the relevant professionals in the decades around 1800 – after which we follow the patterns through to the twentieth century.

Notes

- 1 For first-rate recent accounts of the scientific revolution, see Henry (1997), Schuster (1990) and Shapin (1996).
- 2 I am indebted to the seminal but unpublished PhD by Randall Albury, which showed in 1972 the relations between the elements of chemistry and of medical analysis. See also Albury (1977).
- 3 For an accessible and authoritative history of chemistry, and for full further references, see Brock (1992).
- 4 For a survey, see Dhombres (1989).
- 5 Faraday, Magnus, Regnault and G. C. Foster (of University College, London) might start the list; and Helmholtz moved to physics from physiology. See also Nye (1996).
- 6 See the very useful article by Crosbie Smith (1990).
- 7 On rationalised production in France and the US military see Alder (1997) and Smith (1985). On scientific management Cohen (1997).
- 8 For agricultural rationalisation see Thompson (1978).
- 9 I am indebted here to the work of Isabel Phillips (1989) on styles of animal breeding in the nineteenth century.
- 10 For more on Smith and his fellow chemists, see Farrar (1997) and Gibson and Farrar (1974).